

Secure Wireless Aircraft Data Load

DESIGN DOCUMENT

Team Number: 36

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Executive Summary

Collins Aerospace is requesting that we build a prototype to test and verify the secure wireless upload of data to an aircraft. Wireless security is one of the biggest security concerns in the aircraft industry. Wireless transmission is feasible in that it allows for data to be transferred without the hassle of obtaining and utilizing hardware and cabling to handle data transfers. The downside is that there are attacks and methods of exploitation that are common in the wireless segment of cybersecurity. The confidentiality and integrity of data in transmission are important to ensure that individuals cannot access, view, or tamper with sensitive information. Our objective is to transfer data wirelessly and safely from a hardware controller to an aircraft.

DESIGN REQUIREMENTS

- Follow the industry and Collins Aerospace's own security protocols/standards.
- Simulate a realistic testing environment for simulated data uploads and cyber attacks
- Send a variety of data(e.g. Software updates, topographical maps, in-flight media) to the aircraft.

TECHNOLOGIES

- Two Raspberry Pi 5 microcontrollers to represent the sending of information and receiving intermediary components of full payload transmission.
- Laptops to develop software to load onto Raspberry Pi's.
- Laptop or another Pi to simulate cyber attacks.
- Open source ARINC 615A Aircraft Data Loader code.

PROGRESS UPDATE

- We have decided on using Raspberry Pi's as our testing hardware.
- A basic workflow has been designed and documented to use as a baseline for setting up a testing environment.
- A list of standards, algorithms, and protocols has been made to adhere to Collins' Aerospace and industry standards.

DESIGN NEEDS

- Our initial general design has been approved by our client. The workflow is realistic and can be set up without issue. As time goes on, we will dig deeper into the details and add components to the design.

POTENTIAL NEXT STEPS

- Obtain hardware and develop open source software to meet our requirements.
- Continue to research and design a prototype and architectural structure to accomplish our goal.
- Talk through the general workflow and discuss gaps in efficiency.
- Analyze the open source code.

Learning Summary

DEVELOPMENT STANDARDS & PRACTICES USED

Hardware:

- Raspberry Pi 5
- Personal Computers
- Ethernet for configuration

Software:

- ARINC 615A
- EAP-TLS security protocol
- WPA2 Wifi Configuration

APPLICABLE COURSES FROM IOWA STATE UNIVERSITY CURRICULUM

- CprE/CybE 3310 - Application of Cryptographic Concepts to Cyber Security
- CPR E 4370 - Wireless Security
- CPR E 431- Basics of Information Security
- CPR E 4360- Digital Forensics
- COM S 2520- Linux Fundamentals

NEW SKILLS/KNOWLEDGE ACQUIRED

List all new skills/knowledge that your team acquired which was not part of your Iowa State curriculum in order to complete this project.

NEW KNOWLEDGE

- The avionic security protocols for data transfers that Collins Aerospace follows
 - ARINC 615A
 - ARINC 645
 - ATA SPEC 42
- EAP-TLS verification
- Traditional(wired) aircraft data load protocol

LIST OF FIGURES/TABLES/SYMBOLS/DEFINITIONS

- Fig. 1.1 blahblahblah
- Fig.1.2 blahblahblah
- Fig 2.1 blah blah blah
- ...
- Fig 2.10 blah blah blah

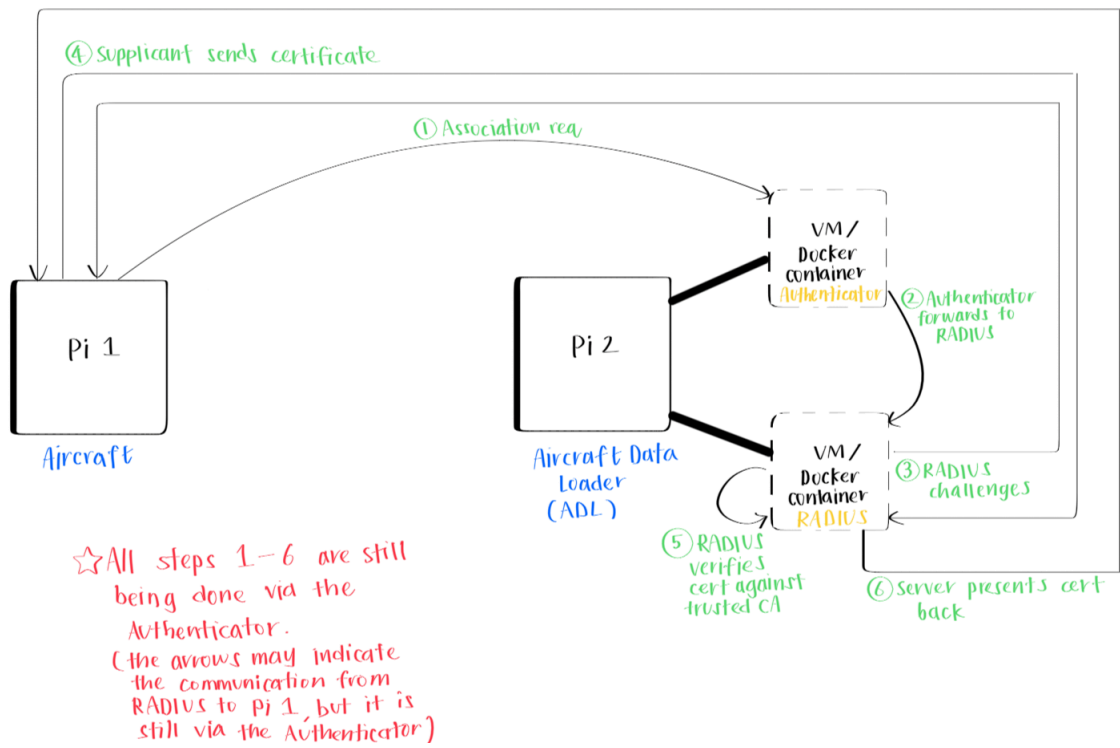


Fig x.

1. Introduction

1.1. PROBLEM STATEMENT

Every modern commercial aircraft is essentially a flying computer system that requires frequent software updates to maintain safety, add new features, and fix bugs, similar to a smartphone or laptop. However, unlike the Wireless Over-The-Air (OTA) updates we commonly see in our daily lives, updating aircraft software today requires an understanding of how to efficiently and securely transmit the data at a higher, more complex level.

Currently, maintenance technicians physically connect laptops to the aircraft using cables and wires, which is a tedious process that can sometimes take hours. Additionally, the plane must be in an idle position and close to the hangar for the wires to reach. In an industry where every minute on the ground can cost airline companies money and delay frustrated passengers, moving forward from this outdated process is crucial to prevent the significant bottleneck caused by physical connections.

In terms of security, aircrafts are becoming increasingly more connected, and cyber threats are growing more sophisticated. The aviation industry faces a lot of pressure to modernize its data loading procedures while maintaining the highest security standards available. After all, we are talking about systems that keep hundreds of passengers safe at high altitudes; any compromise to such a system could have life-threatening effects for the people on board the aircraft.

Our project tackles this challenge by developing a secure wireless data loading system that can change how Collins Aerospace maintains its fleets. This solution aims to reduce aircraft downtime due to slow and inconvenient physical updates, ease the burden of maintenance crews, and pave the way for more efficient airline operations. By creating a prototype that balances the convenience of wireless technology with robust security and certification requirements of commercial aviation, we are working to enable a solution that could change the future of aviation.

1.2. INTENDED USERS

Our secure wireless data loading infrastructure is built to benefit a wide range of uses within the aviation industry. By shifting away from cumbersome physical updates, we create value for the people on the front lines of aircraft maintenance, the airlines managing complex operations, and the passengers who rely on safe airline travel.

Aircraft Maintenance Technician(s)

- **Description:** John is a maintenance technician for an aircraft. He works in a fast-paced environment and is typically under tight time constraints. His primary responsibility is ensuring that the aircraft is safe and airworthy.
- **Needs:** John needs to perform critical software updates quickly without being tethered to the aircraft by a cable. He needs a process that minimizes the time and risk of errors during the data loading process. In addition, John has many aircraft to attend to, so a physical connection with each aircraft would take too long. Above all, he needs to ensure that the software being installed has not been tampered with, because the safety of the aircraft and passengers depends on it.
- **Benefit and Value:** Our infrastructure directly addresses John's needs by eliminating the hours-long process of a physical connection. With a secure wireless transfer, he can initiate

an update from a comfortable location inside the airport. Since John can load the updates wirelessly, he can update all the aircraft much faster and without being directly on the aircraft.

Pilot

- **Description:** Captain Thomas is an aircraft pilot with the main concern being the safety of the passengers aboard.
- **Needs:** Assurance that the software update(s) are complete to ensure passenger safety. He needs to ensure that the process is secure against current cyber threats.
- **Benefit and Value:** He also benefits from on-time departures. The secure and verifiable nature of the wireless data load means he can trust that the software on his aircraft is authentic and has been installed using a certified, tamper-proof process.

Passengers

- **Description:** Jim is a passenger on an aircraft. He is expecting a smooth flight with entertainment provided. Their safety is a top priority.
- **Needs:** The passengers need in-flight entertainment as it was provided with their ticket, and they paid for it. The passengers also need things such as Wi-Fi and weather updates. These passengers are busy and expecting to arrive at their destination on time with minimal delays.
- **Benefit and Value:** The solution we are creating aims to make the passenger experience easier than it already is. The wireless transmission will be able to load entertainment and other vital data onto the aircraft in a quick and efficient manner.

2. Requirements, Constraints, And Standards

2.1. REQUIREMENTS & CONSTRAINTS

List all requirements for your project. Separate your requirements by type, which may include functional requirements (specification), resource requirements, physical requirements, aesthetic requirements, user experiential requirements, economic/market requirements, environmental requirements, UI requirements, and any others relevant to your project. When a requirement is also a quantitative constraint, either separate it into a list of constraints or annotate at the end of the requirement as “(**constraint**).” Ensure your individual requirements are realistic, specific, and reflective or in support of user needs, and ensure your collective requirements are comprehensive.

Functional requirements:

- Maximum load size = 2 TB
- File transfer time window = NA (loads happen in pieces between flight legs)
- Base software image = Linux
- Aircraft and Data Loader must authenticate each other
- Load data from supplicant to server using two Raspberry Pi's
- Simulate Weight On Wheels/Load Ready check

- ARINC 615A
 - Follow Trivial File Transfer Protocol
 - Data Loader gets the aircraft's software configuration with the ARINC 615A Information operation
 - Data Loader transfers loadable software parts to the aircraft with the ARINC 615A Upload operation
- ARINC 665
 - Defines Loadable Software Parts (LSPs) and a Software Directory (LSD)
 - Acceptable ARINC 665 Packages contain:
 - Valid LSD file (directory of all parts, metadata)
 - All required LSPs with correct naming and versioning
 - CRC checksums
 - Overall, ARINC 665 is a standardized format for packaging software into LSPs and organizing them with a LSD.

Physical/Aesthetic Requirements:

- 2 Raspberry Pi Version 5 Microcontrollers
- Mini SD cards for each Raspberry Pi for storage
- Laptops for connecting to microcontrollers and writing code
- Maximum distance from target = 250 ft (length of a 747)

2.2. ENGINEERING STANDARDS

What Engineering standards are likely to apply to your project? Some standards might be built into your requirements (e.g., many projects use 802.11 ac wifi standard) and many others might fall out of design.

Q1) Browse the videos available on the IEEE Standards University site

(<https://www.standardsuniversity.org/videos/>/(Links to an external site.)). Select and watch the following

video: <https://www.standardsuniversity.org/video/standards-education-young-professional/> / (Links to an external site.), as well as 2 other videos of your own choosing. Note that these videos are generally only 2-3 minutes long.

Engineering standards are increasingly more important as time goes on. These standards provide a consistent framework for designing and manufacturing components and systems. They ensure safety, reliability, and efficiency in products and processes. With the use of standards, errors and legal risks are reduced. All in all, standards provide a foundation for safety, performance, and regulatory compliance in engineering.

Q2) Review each of the selected standards. These can be quite lengthy documents, and you are not expected to read through them in their entirety.

Describe, entirely in your own words, what each standard is about and what it is intended to accomplish.

[IEEE 802.11ax: An Overview of High-Efficiency Wi-Fi \(Wi-Fi 6\)](#)

A little overview of this standard: Wi-Fi 6 is an improvement from its predecessors Wi-Fi 5 and Wi-Fi 4. Wi-Fi 5 and Wi-Fi 4 were focused on reaching higher peak data rates. Wi-Fi 6 prioritizes better overall network efficiency. IEEE 802.11x is relevant to our project because it fundamentally addresses the challenges of reliable and secure communication. There are key features that make it applicable to our situation. The first is that it has longer OFDM symbols (12.8 us and 3.2 us), which provide great multipath resistance against metal reflections from aircraft surfaces. The second is BSS coloring with spatial reuse, and this allows the data loading system to operate effectively by distinguishing between networks and ignoring irrelevant interferences. The third is mandatory WPA3 enterprise security with protected management frames: this provides aerospace-grade encryption and prevents deauth attacks which is critical for secure data transfers. The 4th is 1024-QAM modulation and 80/160 MHz channels for throughput of 600+ Mbps, and this allows large database loads to complete in seconds. Lastly is the improved PHY efficiency with OFDMA, and this ensures consistent performance even when channel conditions vary during the transfer. The main aspect that is relevant to our project is implementing WPA3 to encrypt the data load and maintain confidentiality in case of interception along the transfer path.

IEEE Software and Systems Engineering Standards Used in Aerospace and Defense

The importance of this engineering standard is regarding aerospace and defense software standards that focus on life-cycle processes, V&V, testing, and DevOps practices. The key standards that are covered are IEEE 15288, which focuses on life cycle processes applicable to meeting supplier needs promptly. The IEEE 12207 standard focuses on software life cycles to promote incremental improvements for technological advancements. The IEEE 29119 and IEEE 1012 standards focus on software testing and verification to ensure the software is tested and documented properly so that it can be verified for deployment into hardware and software. These standards are crucial for assuring safety in the aerospace industry to make sure the product is well documented, heavily tested, and verified. It is very important to make sure you follow the set guideline standards to reach the maximum level of quality for a given product.

Software & Hardware Configuration Management in Systems Engineering

The importance of this engineering standard is focusing on the configuration of software and hardware systems, and outlines the minimum requirements that need to be met for releasing engineering products into the market. The main standard that is focused on is IEEE 828 and extends into modern practices such as Agile SCM, supply chain risk, SBOM, and organizational configuration management assessments. This standard highlights the importance of the identification of configured items, the status reporting, and the release of engineering products. IEEE 828 is the industry baseline for governing what configuration management a program must have, giving aerospace/defense projects the essential requirements needed to deliver a secure system.

Q3) After reviewing some of the technical details of the three published standards, do you believe it to have relevance to your project? Why or why not? Be specific.

Yes, these requirements are necessary for our project to make sure we follow all the protocols for building a secure system. We are working with aircraft, and it is important to ensure the safety of the passengers of the aircraft by following protocols. Protocols ensuring efficient software life-cycles and promoting validation testing are very important to follow. It is also very important to follow

these guidelines because then your software is interchangeable and easier to maintain. Each one of these standards has the underlying principle to make sure the product is well documented, the information is secure, and the data is being transmitted properly.

Q4) Review with your team the standards that each of you have selected. What other standards did some of your team members choose that are different?

Our client has identified several standards pertaining to avionic software, security, and authentication. It is important that we follow these standards since our solution will be used in a professional environment, and must comply with the rest of the industry.

Additional Standards:

- EAP-TLS
- ARINC 615A
- ARINC 665
- NIST 2.0

Q5) What modifications do you intend to make to your project design to incorporate these standards?

We have taken into account most of these standards to already be incorporated into our project. Some of them may need to be looked into a little more, such as NIST 2.0, mainly because we need to be certain that we have a clear understanding of the requirements from this standard that may apply to our project.

3. Project Plan

3.1 PROJECT MANAGEMENT/TRACKING PROCEDURES

What project management style are you adopting (e.g., agile, waterfall, or waterfall+agile)? Justify your project management style with respect to the project goals and contexts.

Also, describe how your team will track progress throughout the course of this and the next semester. This could include Git, Github, Trello, Slack or any other tools helpful in project management.

Management Style:

We are adopting an Agile Project Management style. This is due to the evolving nature of the requirement and the constant need for client feedback. Agile is best suited for rapidly iterating on hardware and software integration and responding quickly to any cybersecurity developments or issues discovered during simulation and testing.

Justification:

- Frequent interactions with Daniel Riechers (Collins Aerospace) support continual feedback and priority changes, minimizing the risk of building unnecessary features and setbacks.

- A complex and staged environment, including hardware setup, network configuration, and cyberattack simulation, lends itself to being broken down into sprints.
- Each sprint delivers verifiable test increments (i.e., hardware communication, TLS implementations, and cyber attack scenarios).

Tracking Progress:

- **Gitlab:** For version control and tracking software/code changes as requirements shift.
- **Google Drive:** For documentation, design updates, and tracking time.
- **Microsoft Teams:** For meeting notes and meeting schedule.

Team members will update tasks daily and participate in weekly review meetings to demonstrate sprint deliverables and adjust priorities as needed.

3.2 TASK DECOMPOSITION

In order to solve the problem at hand, it helps to decompose it into multiple tasks and subtasks and to understand interdependence among tasks. This step might be useful even if you adopt agile methodology. If you are agile, you can also provide a linear progression of completed requirements aligned with your sprints for the entire project.

- **A: Requirements Gathering**
- **B: Design and Planning**
- **C: Hardware Acquisition and Setup**
 - Order 3 Raspberry Pi 5 (8 GB) units
 - Prepare SD cards, cooling cases, and laptops
 - Physical network setup for controller and aircraft simulators
- **D: Environment Configuration**
 - Configure ethernet and Wi-Fi networks for secure baseline
 - Implement testing and performance benchmarks
- **E: Software Development**
 - Port and validate open source ARINC 615A Loader onto Raspberry Pi platforms?
 - Develop secure Wi-Fi implementation (WPA2, EAP-TLS)
 - Simulate data file preparation with compliance to ARINC 665 (SW update, list of parts, media bundles)
- **F: Standards & Protocol Compliance**
 - Align implementation with ARINC 615A, ARINC 665, EAP-TLS, IEEE.....
 - Document compliance verification checklist
- **G: Cyberattack Simulation**
 - Configure a third device (laptop or raspberry pi) for cyberattack purposes
 - Develop and launch various attack scenarios: ie., deauth attacks, packet interception, man-in-the-middle, evil twin, and replay attacks
- **H: Security Hardening**
 - Integrate CRC checking, authentication protocols
 - Test encryption/decryption procedures and certificate rotation
- **I: Testing and Evaluation**
 - Conduct unit, interface, integration, and system tests on hardware-software setup

- Perform acceptance and security testing for all functional requirements
- **J: Documentation and Reporting**
 - Prepare design documentation, standard operation procedures, and test reports
 - Present findings, gaps, and next steps to client in regular review sessions

3.3 PROJECT PROPOSED MILESTONES, METRICS, AND EVALUATION CRITERIA

What are some key milestones in your proposed project? It may be helpful to develop these milestones for each task and subtask from 3.2. How do you measure progress on a given task? These metrics, preferably quantifiable, should be developed for each task. The milestones should be stated in terms of these metrics: Machine learning algorithm XYZ will classify with 80% accuracy; the pattern recognition logic on FPGA will recognize a pattern every 1 ms (at 1K patterns/sec throughput). ML accuracy target might go up to 90% from 80%.

In an agile development process, these milestones can be refined with successive iterations/sprints (perhaps a subset of your requirements applicable to those sprints).

Milestone	Metric/Evaluation Criteria	Proposed Timeline
Order 3 Raspberry Pi 5's, 8 GB of RAM for each	Consulted with Prof. Selim and Mr. Riechers about our order, order HW with their approval.	10/12-10/19
Initial Pi2Pi connection	Able to make basic communication between 2 pis over Wifi	10/20-10/27
Create and test an access point on Raspberry Pi	Initial connection from pi to pi is established	10/28-11/4
Basic Data load from Pi to Pi	Able to load data from 1 pi to another	11/5-11/12
Test Security of transfer	Try to access data in transfer from an attack machine. Successful if unable to retrieve any data in transit.	11/13-11/20

Fig x. Proposed milestones and accompanying timeline.

3.4 PROJECT TIMELINE/SCHEDULE

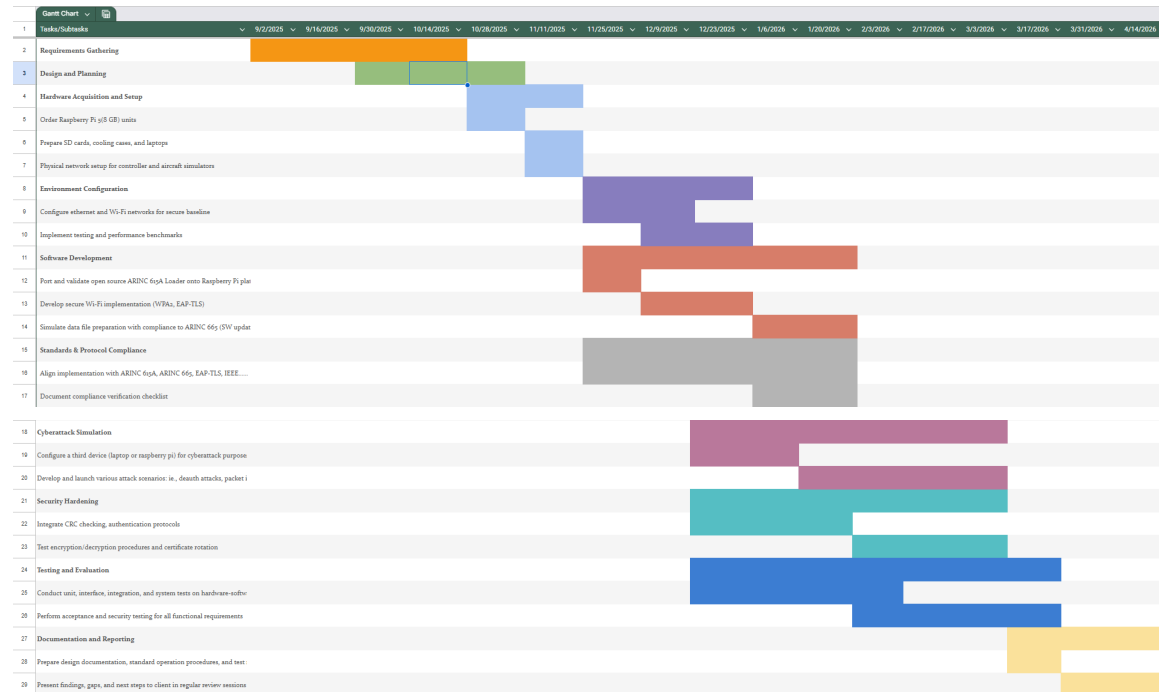


Fig x. Gantt chart for project timeline.

3.5 RISKS AND RISK MANAGEMENT/MITIGATION

For each task, identify all salient risks (certain performance targets may not be met; certain tool may not work as expected) and assign an educated guess of probability for that risk. For any risk factor with a probability exceeding 0.5 and each high severity risk, develop a risk mitigation plan. Can you eliminate that task and add another task or set of tasks that might cost more? Can you buy something off-the-shelf from the market to achieve that functionality? Can you try an alternative tool, technology, algorithm, or board?

Based on our research conducted so far, we have determined that using Raspberry Pi's is the best possible solution to our problem. While there may be other existing solutions such as USRP's, there are multiple reasons why we ended up going with Raspberry Pi's.

Tasks and their risks:

Task	Possible Risk	Likelihood	Mitigation Strategy
Requirements Gathering	Missing or ambiguous stakeholder requirements lead to scope creep and rework.	0.5	Maintain requirement traceability, and perform peer reviews early.

Design and Planning	Architecture may not meet latency or security constraints, requiring redesign.	0.5	Prototype early; conduct design reviews with domain experts.
Hardware Acquisition and Setup	Delays or incompatible hardware stall schedule.	0.7	Identify multiple suppliers; verify specs before ordering.
Order Raspberry Pi 5 (8 GB) units	Parts shortages or counterfeit units cause unusable devices.	0.5	Order early from authorized vendors and test one sample batch.
Prepare SD cards, cooling cases, and laptops	Image corruption or cooling issues cause instability.	0.5	Verify all images; stress-test hardware before deployment.
Physical network setup for controller and simulators	Incorrect VLAN/subnet setup causes connectivity issues.	0.5	Use documented network diagrams and peer verification.
Environment Configuration	Inconsistent setup between devices causes non-reproducible bugs.	0.7	Use automation scripts and configuration management tools.
Configure ethernet and Wi-Fi networks for secure baseline	Misconfigured security prevents access or introduces vulnerability.	0.5	Validate network configs using checklists and security scans.
Implement testing and performance benchmarks	Unrealistic benchmarks give false performance confidence.	0.5	Define test scenarios with real-world loads; validate metrics.
Software Development	Dependency or API incompatibilities introduce vulnerabilities.	0.5	Use version control, CI/CD, and dependency scanning tools.
Port and validate ARINC 615A Loader	Porting introduces timing or IO issues that break protocol compliance.	0.5	Incremental testing on each Raspberry Pi build; use reference data.
Develop secure Wi-Fi (WPA2, EAP-TLS)	Certificate or config errors expose credentials.	0.5	Implement certificate validation automation; conduct penetration tests.

Simulate data file preparation (ARINC 665)	Non-compliant files cause software update failures.	0.5	Validate file format using schema checks and reference data sets.
Standards & Protocol Compliance	Misinterpretation of specs leads to incomplete compliance.	0.5	Conduct joint reviews with avionics compliance experts.
Align with ARINC 615A/665/EAP-TLS/IEEE	Conflicts between standards cause implementation gaps.	0.5	Map overlapping requirements and resolve conflicts in design phase.
Document compliance verification checklist	Incomplete checklist misses verification steps.	0.1	
Cyberattack Simulation	Simulated attacks disrupt non-test systems.	0.5	Isolate network environment; obtain approval before simulation.
Configure third device for cyberattack	Poor isolation allows unintended lateral movement.	0.5	Use VLAN segmentation and sandboxing for attack machines.
Develop and launch attack scenarios	Legal/privacy issues if executed improperly.	0.1	
Security Hardening	Hardening breaks functionality or complicates maintenance.	0.5	Apply changes incrementally; test after each security enhancement.
Integrate CRC checking & authentication	Faulty implementation corrupts legitimate updates.	0.5	Validate CRC routines using known-good test vectors.
Test encryption/decryption & certificate rotation	Key management mistakes cause outages.	0.5	Automate certificate renewal; maintain recovery keys offline.
Testing and Evaluation	Test coverage gaps miss critical defects.	0.5	Develop a formal test matrix; automate regression testing.
Conduct unit/integration/system tests	Hardware variability causes intermittent failures.	0.5	Test across multiple units; document test environments.

Perform acceptance & security testing	Misunderstood criteria lead to rejection.	0.1	
Documentation and Reporting	Documentation outdated or incomplete.	0.5	Version control all documentation; assign review ownership.
Prepare design docs, SOPs, test reports	Poor organization hinders handover.	0.1	
Present findings & next steps	Poor communication causes misalignment.	0.1	

Fig x. TaskxRisk management chart.

3.6 PERSONNEL EFFORT REQUIREMENTS

Include a detailed estimate in the form of a table accompanied by a textual reference and explanation. This estimate shall be done on a task-by-task basis and should be the projected effort in the total number of person-hours required to perform the task.

Task	Estimated Person-Hours
Hardware Setup	24
Network Configuration	16
Software Development	40
Protocol/Standards Review	20
Cyberattack Simulation	28
Security Hardening	24
Testing & Evaluation	40
Documentation/Reporting	22

Fig x. Estimate of personal effort requirements.

3.7 OTHER RESOURCE REQUIREMENTS

Identify the other resources aside from financial (such as parts and materials) that will be required to complete the project.

- Hardware:
 - 3 8 GB Raspberry Pi 5 units
 - SD cards, cooling cases, network cables
- Software:

- ARINC 615A open source loader code, EAP-TLS libraries
 - Wi-Fi security suite, cyberattack scripts/tools
- Development Machines:
 - Team member laptops, spare Pi for attack simulation, access to client environments for integration tests
- Networking:
 - Wireless routers/access points, Ethernet switches for fallback/data validation
- Testing Data:
 - Sample software updates, topographical maps, media files for load simulation
- Documentation:
 - Templates, standards checklists, reporting frameworks
- Other:
 - Access to Collins Aerospace documentation, industry standards libraries

4 Design

4.1 DESIGN CONTEXT

4.1.1 Broader Context

Area	Description	Examples
Public health, safety, and welfare	The wireless data-loading system directly affects the safety of aircraft operations by ensuring software updates and securely and accurately transmitted. It is crucial to make sure the data is being transmitted properly to <u>guaranteed the safety of the passengers.</u>	Enhances passenger safety and crew safety through following ARINC security clearances and other validations.
Global, cultural, and social	Air travel is a global industry that demands safety and reliability. The project aligns with international aviation standards and supports the global needs for efficient improvement of cyber-secure aircraft maintenance and data transferring	Supports worldwide airlines and maintenance crews for all Collins Aerospace clients.
Environmental	Wireless data loading reduces the need for physical cabling, hardware, and frequent equipment replacing, resulting in lowering e-waste.	Less cable waste; lower hangar power usage; extended equipment lifespan; reducing carbon footprint.
Economic	The projet saves time and money by streamlining aircraft software updates and inflight entertainment. It also enables a central location for data transferring which lowers cost for additional hardware.	potential commercialization for this product if it is effective and can result in profits for Collins Aerospace.

Fig x. Table of relevant considerations related to our project in following areas.

4.1.2 Prior Work/Solutions

There is no solution currently in place that is attempting to address what we are (wireless dataloading). However, we have heard from our client advisor that Collins has a wired solution in place for dataload, but not a wireless one, and that is about all we got. These wired solutions are currently in place in many aircraft and require operators to manually plug in wires during dataload. Our solution aims to remove this human interference and load the data wirelessly to each device without requiring any plug-ins. Our solution will utilize hardware such as Raspberry Pi's and an access point along with EAP-TLS and ARINC 615 to possibly create a new industry standard when it comes to aircraft dataloading.

EAP-TLS is a major product that we will be working very closely with. EAP-TLS is a wireless communications protocol and forms the basis for wireless security in this deployment. More specifically, the EAP-TLS authentication is based on two-way authentication between the client and RADIUS authentication server, rather than between AP and Authenticator [3]. This protects against unauthorized access and interception. This protocol allows us to establish a safe and reliable connection between the access point and the devices. This is currently the industry standard in wireless communications within avionics and will be a critical security component within our solution.

2 major products we will be working closely with are Raspberry Pi's and an Access Point. The access point we are looking into is a Cisco 240ac. We needed to find an access point that supported WiFi 5 and WPA2/3 in order to properly implement EAP-TLS. To start, this access point uses 802.11X, RADIUS for authentication, and has the potential to support up to 400 clients [2] which directly aligns with our project and should be a seamless option. On top of this, it has a fairly good throughput, is decent in price (\$150), supports WiFi 5, and it would be a good fit for our needs. This access point will be able to take information from the dataloader and be able to transmit it to the target. Additionally, by nature, this access point supports EAP-TLS integration. Next, we have the Raspberry Pi's. We have one Raspberry Pi which will specifically mimic a dataloader. In this scenario it is going to have the ARINC 615A protocol on it. The second Raspberry Pi will be a target to receive the data. Both of these are also easy to incorporate into the EAP-TLS framework.

ARINC 615A is an aviation data loading standard widely adopted across the avionics industry for secure software and data transfer in modern aircraft. It has been thoroughly verified and certified with hours in-use and in-service [1]. ARINC 615A specifies the technologies and methods in place for current and future dataloading in avionics. It leverages networking protocols such as WiFi data loading, which fits right into our project guidelines. As stated earlier, ARINC 615A is an industry standard, this means that if we want our solution to seamlessly be implemented across the industry, this will need to be followed. Overall, this standard represents what will need to be done for wireless communication, and if we want to build a complete solution we will need to follow this standard closely.

Comparison (Current Wired Solution vs Proposed Wireless Solution)

Solution	Pros	Cons
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Current Wired Dataloading Solution	Certified, verified, dependable, industry standard, fast, standardized and currently deployed.	Requires extra work, operates need to be physical proximity to load data, high downtime and requires a lot of cable management
Proposed Wireless Solution (ARINC 615A w/ EAP-TLS) for secure data loading	Less cables are required, operators no longer need to be in physical proximity or plug anything in, it is scalable, would require much less overhead to operate	Requires a lot of integration with existing standards and protocols, will be slower in data speeds than a wired solution, requires all new devices and components so will be costly to implement across the industry

Fig x. Traditional wired vs. proposed wireless solution pros and cons.

[1] Performance Software, "What is ARINC-615A and Why it Matters," Performance Software, Sep. 20, 2022. [Online]. Available: <https://www.psware.com/what-is-arinc-615a-and-why-it-matters/> [Accessed Oct. 28, 2025]

[2] Cisco, "Cisco Business 240AC Access Point Data Sheet," Cisco, Mar. 27, 2020. [Online]. Available: <https://www.cisco.com/c/en/us/products/collateral/wireless/business-200-series-access-points/smb-01-bus-240ac-ap-ds-cte-en.html>. [Accessed: Oct. 28, 2025]

[3] W. Huang and R. Li, "WLAN Authentication System Based on the Improved EAP-TLS Protocol," 2009 Second Pacific-Asia Conference on Web Mining and Web-based Application, Wuhan, China, 2009, pp. 181-184. [Online]. Available: <https://ieeexplore.ieee.org/document/5232480>. [Accessed: Oct. 28, 2025]

4.1.3 Technical Complexity

Subsystems/Principles:

- Wireless Security
 - Our project implements EAP-TLS authentication over WPA2-Enterprise, providing mutual authentication between aircraft and the data loader. This subsystem uses digital certificates and TLS encryption to ensure that only trusted devices can exchange data.
- Data protocol
 - The data protocol subsystem follows the ARINC 615A standards, which governs how avionic systems load and transmit data. It operates over Ethernet/TFTP to manage sessions establishment, file segmentation, and acknowledgement of loadable parts between the data loader and the aircraft.
- Data Integrity
 - Ensuring integrity and correctness of transferred software is achieved through the ARINC 665 standard. This subsystem manages how Loadable Software Parts (LSPs)

are packaged, referenced, and validated using Cyclic Redundancy Checks (CRC) and a Loadable Software Directory (LSD). These verification steps guarantee that the received software is untampered and consistent with its expected configuration.

- Cybersecurity Simulation
 - To validate system security, a separate testbed simulate real-world cyberattacks such as deauthentication, replay, and man-in-the-middle attacks. This subsystem allows evaluation of the robustness of EAP-TLS and Wi-Fi 5 security under adversarial conditions.
- Embedded Hardware
 - The embedded subsystem consists of two Raspberry Pi 5 devices acting as the aircraft data loader and client transmitter. As well as one CISCO wireless Data Access Point used to coordinate these devices demands of computing resources, real-time I/O, and network synchronization. Each unit must handle simultaneous encryption, data transmission, and integrity checks without performance degradation.
- Compliance
 - The compliance subsystem ensures that the project aligns with the relevant industry standards including IEEE 802.1X, ARINC 615A, ARINC 665, EAP-TLS, NIST 2.0, and IEEE 828 configuration management guidelines. These frameworks govern safety and lifecycle documentation across aerospace systems.

4.2 DESIGN EXPLORATION

4.2.1 Design Decisions

- Number of Pis: We toyed with one, two, and three Pi designs before deciding on 2 Pis. We could simulate a testing environment for our solution using any of those options, but we decided that the two Pi solution is most similar to the real world application. One Pi acts as the data loader in the plane, the other Pi acts as any number of load targets. This could influence project success greatly because it is a large budgeting decision and totally changes our high level design.
- Authentication hosted on VM: From the start of our project the client has made it clear that we will have to implement EAP-TLS for our authentication process. A proper EAP-TLS handshake includes three machines(Supplicant, FreeRADIUS Server, Authenticator). In our design, the Server and Authenticator are hosted on Virtual Machines because that is how it would usually be done in real life. The supplicant is just whatever is trying to connect to the Server, so the supplicant is our Target(one of the Pis). This was one of the first design decisions we made, thus our design is heavily influenced by this. It is the first step in all our data transfers so it is crucial to get it right.
- Raspberry Pi Version/Size: Raspberry Pi 5 is the most recent version of the device. More specifically, Raspberry Pi 5 16 GB is the most recent version, but it comes in sizes 2, 4, and 8 GB as well. So we had to consider the level of computing we need to do in our project and decide which device was the best fit. From our research, 4 GB would most likely have worked as the minimum amount of RAM, but 8 GB would give us better performance. 16 GB was total overkill(used for big data computations and AI model training). This choice

was important because it could make or break our budget, but the actual performance isn't as important since we will be comparing it to actual field values anyways.

4.2.2 Ideation

We started with a team brainstorming session where we listed what we needed: hardware that could handle encryption, had good networking, fit our budget, and could be programmed easily. We researched the single board computers and embedded systems online, looking at datasheets and specs which were thankfully all publicly available to download and read, some of the community forums to see what problems could arise, and then what other projects were created for similar applications.

5 Options:

1. Raspberry Pi 5. It is the latest Pi with hardware encryption gigabit ethernet, 8GB RAM option, strong community support, extremely strong encryption mechanisms, and was around 70 to 150 dollars
2. Raspberry pi 4. Previous generation, cheaper, but not any hardware crypto acceleration which means our data transfers would be really slow
3. Arduino-based systems - Super cheap but they are limited in processing power and networking capabilities.
4. BeagleBone Black - This is somewhat popular and it's pretty decent for embedded systems. It has more gpio pins but the processor is fairly weak.
5. NVIDIA Jetson Nano - Ok this one was a stretch but it has an insanely powerful GPU and its around 150 dollars. It is definitely overkill for our needs and we don't need any AI processing. Not to mention it lacks hardware crypto acceleration.

Raspberry Pi 5 had everything we needed so we didn't have much to consider when deciding. Hardware crypto acceleration is the most important aspect of this project.

4.2.3 Decision-Making and Trade-Off

Criteria	Weight	Pi 5	Pi 4	Arduino	BeagleBone	Jetson Nano
Encryption Speed	25%	5	2	1	2	3
Network Performance	25%	5	4	2	3	4
cost	20%	4	5	5	4	4
Ease of development	15%	5	5	3	3	3
Community support	15%	5	5	4	3	3
Total score	100%	4.7	3.9	2.8	2.9	3.4

Fig x. Weighted trade-off chart for hardware.

Key Tradeoffs:

Encryption speed vs Cost: Pi 4 was cheaper but its lack of hardware crypto would make our secure transfers slow.

Power vs Simplicity - Jetson Nano had more power but we did not need GPU capabilities

Industrial vs Experimental Prototyping - Well the beaglebone is definitely more industry focused no doubt, but weaker specs meant we would struggle to hit data transfer speeds.

Well the Raspberry Pi 5 scored highest because it met all the critical requirements for fast hardware encryption, networking, and affordable enough to be both transmitter and receiver. Fits in the budget as well. Forums say that the raspberry pi 5 is the most reliable and fun to work with therefore we can assume it will be the same experience for us.

4.3 PROPOSED DESIGN

4.3.1 Overview

The current way data is transmitted on airplanes is to connect through a wired connection to a data loader so it can process updates or uploads onto the aircraft systems. Our project design is to implement a wireless connection to transfer data without the utilization of wired connections while ensuring safe and efficient data transfer. The current design is to utilize three hardware devices and make a prototype for wireless transferring on current airplanes. One device is simulating the current wired data loader that loads all the crucial data on an airplane. Another device acts as a base station that receives and loads data wirelessly to the wired device. The third device is the transmitter that sends the important data to the base station, so it can upload all the important information to the airplane.

4.3.2 Detailed Design and Visual(s)

We aim to make a prototype that simulates the implementation of data transferring that is currently on aircraft. Aircrafts conduct data transfer through a wired connection and our client from Collins Aerospace aims to make a prototype to transfer data wirelessly. Our implementation requires a wired hardware base station, such as a CISCO 240AC access point that will simulate data transferring on aircraft, so we can make this project as feasible as possible to how data transferring is currently done. We will have two Raspberry Pis that will conduct the data transfer wirelessly through Wi-Fi connection. One Pi will conduct EAP-TLS, the gold standard of certification security, and will host the DENT-Authenticator and FREE-Radius server to authenticate users who want to transfer data properly. The other Pi will simulate the process of sending data over Wi-Fi to the base station to simulate how maintenance workers will utilize our project implementation.

4.3.3 Functionality

The prototype we are building is to simulate how current employees will conduct software updates and data transferring of in-flight entertainment. An airplane technician will go onto the airplane that needs either software updates or adding new inflight entertainment. The technician will use his computer and connect to the wireless connection. They will be prompted to enter his credentials through an authentication process to be able to connect to the data transmitter. Once they are verified they will have access to our server and be able to complete whatever action is

required for the given aircraft they are working on. Once they have completed the data transfer they will disconnect from the Wi-Fi connection and exit the plane.

4.3.4 Areas of Concern and Development

Design Needs vs. Current Design

Our current design has been cleared by the client to start prototyping, so we know that it meets the most basic requirements. These include EAP-TLS authentication, communication between targets and a data loader via wireless access point, ARINC 615A data load procedure. However, we won't know how well it satisfies the actual needs of the project until we finish a prototype and start testing.

Requirement Type	User Needs	Our Design	Score
Functional	- Maximum file size (Waiting on client) - File transfer time window (Waiting on client) - Aircraft and Data Loader must authenticate each other - Load data from supplicant to server using two Raspberry Pi's - ARINC 615A - ARINC 665	- Maximum file size = 1 TB - File transfer time window(database) < 5 minutes - File transfer time window(LRU) < 2 hours - EAP-TLS Authentication - Load data from supplicant to server using two Raspberry Pi's - ARINC 615A - ARINC 665	Exceeds
Security	- EAP-TLS Authentication - Secure TFTP transfers - Data Confidentiality and Availability	- EAP-TLS Authentication - AES encryption for TFTP transfers - Data Confidentiality and Availability	Satisfactory
Resource	Dependent on open-source ARINC libraries	ARINC 615A library(not tested)	Satisfactory

Physical	• Uses Raspberry Pis to simulate hardware • Uses laptops to simulate cyber attacks and test security/app UI • Useable on a plane	• 2 8GB Raspberry Pi 5s • Commercial off-the-shelf Wireless Access Point • 2 Virtual Machines from ETG • Laptop to simulate attacks	Exceeds
Aesthetic	• Must include GUI that lets users choose load targets	• ARINC 615A library(not tested)	Unsatisfactory
User Experiential	• Ability to connect and load to multiple targets • Load to a grounded plane	• Load to a grounded plane	Unsatisfactory
Environmental	• None	• None	NA

Fig x. User needs and current design comparison.

Primary Concerns

Feasibility:

The biggest concern with our design is that it won't perform well enough to be considered for a real world application. Our solution is to be a prototype Secure Wireless Dataloader that Collins will use to assess the feasibility of the project. Since we work with a limited budget and commercial off the shelf hardware, we know for a fact that our wireless connection will transfer data slower than wired.

Technical Requirements:

Another large concern is meeting the technical requirements of our client. We have had to research many avionic and data transfer protocols in the last few weeks(ARINC 615A, ARINC 645, ARINC 665, ARINC 429, TFTP). We have designed our solution to align with these protocols, but we are unsure if we meet all the functional requirements laid out in this huge set of standards. This is important since our prototype may be the base design for other implementations in the future.

Security Requirements:

A small concern in our design is adding the security features. We are confident in our design's implementation of EAP-TLS(our authentication process), but we still need to develop the other security elements of our design(e.g. AES Encryption method hasn't been added to our diagram).

Requirement Sufficiency:

Part of our job is to compare our data transfer speed, data size limits, and security of the data in/post transfer. However, since we will be assessing the feasibility of moving forward with our prototype in the real-world, we need some benchmarks and requirements to test against. We are fairly confident in our security requirements, but have had issues with obtaining physical requirements(e.g. Timing, data size, data types, etc.).

Immediate Plans

Immediately, we will collect more benchmarks(physical and functional requirements) from either our client or an industry connection. We need to ask our client or professors to connect us with an airline maintenance technician to ask about the current(wired) process. That will help us both with our feasibility and requirement sufficiency concerns. Also, will add security and software design components to our solution. We only have a mid-level diagram right now and we need to flesh out the individual components. More design documents(low-level design, software sequence diagram, data flow diagram, etc.) will help us clarify elements of our project we are unsure of. Finally, we will consult our faculty advisor about security methods to add to the design.

Questions:

- (Professors/client) Can you connect us with an aircraft maintenance technician? (Or how to find/make benchmarks for our project without one)
- (Client) What does our GUI need to do? What should it look like?
- (Faculty advisor) Is our design transferring data securely?
- (Faculty advisor) What stages of the transfer must we encrypt the data?
- (Faculty advisor) Are we following proper network/software/hardware design protocols?
- (Client) Where have other wireless solutions failed?

4.4 TECHNOLOGY CONSIDERATIONS

Strengths:

Raspberry Pi 5 - The main component of our testing environment is the Raspberry Pi 5. The Pi is a microcontroller that functions as a computer, and the purpose of this device is to simulate transmitter to receiver communication. Aspects of the Raspberry Pi 5 that we will apply to our environment setup:

- ARMv8 cryptographic extensions with hardware AES acceleration. If we compare this to the Raspberry Pi 4, the Raspberry Pi 5 has a 45x performance improvement for encryption operations. We want encryption at high speeds because this makes the process for transferring data securely faster. To get more into the technical weeds, for AES-128-CBC performance(which we are considering using), its encryption/decryption speed ranges from 879-1373 MB/s. This is extremely fast.
- Processor : The Cortex-A76 processor which operates at 2.4 GHz produces AES-256-CBC scores around 1144 MB/s so if we are advised to kick the encryption security up a notch, we can do so and not sacrifice speed that much. This also means at these speeds the data pipeline won't be crypto-bottlenecked to an extent.

Using the raspberry pi 5 is a huge upgrade from other microcontrollers because it has hardware encryption built into the processor that makes it extremely fast. The networking is also solid because it contains a gigabit ethernet hitting 940 Mbps consistently, and you can add faster 2.5 gigabit adapters through usb 3.0 ports that each get their own full 5 Gbps bandwidth instead of sharing. The PCIe slot lets you connect super fast NVMe storage which is way more reliable than SD cards for continuous operation. It has built in real time clock that keeps accurate time even when powered off, which is critical for timestamping secure transfers and even audit logs. The whole unit costs around 150 to 200 full equipped so we can afford to deploy multiple units for redundancy. It is already proven in the aerospace industry too. NASA uses Raspberry Pis in spacecraft and satellites and companies use them in industrial settings where reliability matters.

Weaknesses:

The biggest problem is the standard SD card might fail quickly with continuous read/write operations, so using NVMe storage as an option could be the move. It adds an extra 60 dollars to the cost and it would take up the PCIe slot. It does not have a built in TPM, which is a Trusted Platform Module chip, so we might need to add one for another extra 30 dollars if we want a full secure boot protection from physical attacks but our advisor hasn't mentioned anything about that area yet. When the pi is pushing a lot of encrypted data at high speeds, it gets hot and needs active cooling like a fan. We might need to buy a fan add-on for cooling. To be more specific, then Pi 5 is only rated for 0-50 degrees celsius so if Collins needs it to work in extreme temps then we need to look at industrial variants like the Revolution Pi or add custom cooling like I previously mentioned. There is also no built in power redundancy so if power cuts mid transfer then we will lose data unless we add our own UPS or battery backup. For connectivity, while the gigabit ethernet is fast, it is still limited compared to legit industrial systems that can do stuff like 10 gigabit transfers.

Alternatives:

The only other alternative that comes to mind after research that could be a viable option is the NI CompactRIO which is pretty expensive around 3000 dollars. It is industrial grade real time controller with FPGA programming and it is designed for harsh environments. It can survive in temperatures ranging from -40 degrees to 70 degrees celsius. It has deterministic control and Time Sensitive Networking. It is built for aerospace and defense applications. It is extremely reliable and rugged with pretty intensive I/O options through modular C Series cards. The problem is that it costs almost 30 times more than the Pi 5 and it requires expensive licenses and it also be total overkill if we are moving encrypted files from point A to B.

Tradeoffs:

There isn't much to say for tradeoffs. More expensive microcontrollers are obviously better but better isn't necessarily the primary focus. We are trying to simulate an idea and it would be more impressive to work cost efficiently to show that a solution can be created with cost efficient technology.

4.5 DESIGN ANALYSIS

In terms of physical implementation we do not have anything setup with the Raspberry Pi's since they have not came in yet. We were waiting until we got approval from our faculty advisor and client before ordering the raspberry pi 5's. Luckily we have several of the individual components

done. These individual components are modular to putting everything together with the microcontrollers. Radius server for authentication. It was feasible to setup on a virtual machine because it seems like setting up a radius server on linux is easier and faster than windows. Secondly is docker. Setting up the docker engine on ubuntu was quick therefore transferring it to our actual servers will be seamless.

Implications - There might be implications with the size of the data. Encrypting it and transferring large amounts of data is hard. It can be resource intensive and could break in between transfers if it is a lot of data. The thing is that we won't worry about scaling data until we first get data transferring in general. Once the data is able to transfer successfully, we will scale with larger and larger data.

System Design/Architecture:

We have two architecture diagrams that are in the second draft. This means we showed our first draft to our faculty advisor and client and they gave us places to improve or consider. The two diagrams are a data flow and then encryption flow. Data flow is the modular movement from component to component. Encryption flow is every single aspect of encryption that will be done from point A to point B.

5 Testing

Testing is an **extremely** important component of most projects, whether it involves a circuit, a process, power system, or software.

The testing plan should connect the requirements and the design to the adopted test strategy and instruments. In this overarching introduction, give an overview of the testing strategy and your team's overall testing philosophy. Emphasize any unique challenges to testing for your system/design.

In the sections below, describe specific methods for testing. You may include additional types of testing, if applicable to your design. If a particular type of testing is not applicable to your project, you must justify why you are not including it.

When writing your testing planning consider a few guidelines:

- Is our testing plan unique to our project? (It should be)
- Are you testing related to all requirements? For requirements you're not testing (e.g., cost related requirements) can you justify their exclusion?
- Is your testing plan comprehensive?
- When should you be testing? (In most cases, it's early and often, not at the end of the project)

5.1 UNIT TESTING

Philosophy:

Since our project uses libraries with many pre-built tests, the first step of our testing plan is to review the coverage of those tests. Second, we identify gaps we need to cover for our use case and then begin writing unit tests from there. The process is as follows: we will analyze the viability of these tests and create new testing to match our new implementation following ARINC guidelines.

Third, we will need to test any hardware specific functionality/edge-cases that are caused by our Pis. Finally, we add tests to target the security components of our solution. This will include certificate handling, encryption validation, and even simulated cyber attacks. These are last in our plan because basic functionality is needed before we can test security. This philosophy is the basis of our team's test plan.

We have organized our tests into the following critical units, which cover all the individual components of our system/solution. We based our units on the functional requirements and industry standards of our project to get as close as possible to full coverage.

Critical Units:

- Certificate and Authentication(EAP-TLS and cert handling)
- ARINC Protocols (615A, 665, TFTP)
- Encryption and Security(data encryption/decryption and digital signatures)
- Data Validation(CRC checks, integrity, input validation)
- Network Communication (socket operations and packet handling)
- Simulation(load ready state and cyber attack)
- Hardware Specific(Raspberry Pis and Cisco Wireless Access Point)
- Error Handling

Tools:

Google Test(gtest): C++ testing framework that can integrate with CI/CD pipelines. We will use gtest to automate the majority of our unit testing, since our codebase is solely C++.

CTest: Built-in ctest unit tests allow us to make sure all pre-existing functionality still works as expected. Will be crucial in testing our ARINC Protocols unit.

Wireshark/tshark: Capture packets directly from devices. Allows us to test authentication, certificates, and network communication at a low level.

5.2 INTERFACE TESTING

What are the interfaces in your design? Discuss how the composition of two or more units (interfaces) are being tested. Tools?

Philosophy:

The interfaces in our design is the ARINC615A that connects the TFTP transfer of client information. The TFTP interface can connect to the network socket that the server runs on. The server is connected via EAP-TLS that acts as an interface to help verify the client and build certificates for authentication.

Tools:

Tools such as gtest and gmock can be used to test the interface connections between clients and servers.

Google Test(gtest):

gmock:

5.3 INTEGRATION TESTING

The main critical path is successful mutual trust between the data loader PI and the radius server(auth server). It is crucial to ensure that certificate generated/signed by CA for data loading PI is accepted/recognized by radius server and that the certificate generated/signed by CA for the auth radius is accepted by data loader PI. This is also a path for security testing but its part of integration testing since the functional aspect outside of the security aspect is that if this fails, the data will not be sent. The whole point of this project is for the data to be sent.

Testing Methodology:

1. On any machine, manually create a simple CA.
2. Create Radius Server certificate
3. Create PI client certificate
4. Configure both data loader pi and auth server to function
5. Run integration test from command line and look for "SUCCESS" response

Philosophy:

Integration of ARINC open source libraries is critical. We must make sure that our network and security components, as well as any added software functionality, integrate smoothly with the open source libraries. Component to component testing contributes to successful system integration. If security fails then the data is not sent, therefore ensuring security functions are operational is crucial for integration.

Tools:

Freeradius

Freeradius-utils

Wpasupplicant

eapol_test?

5.4 SYSTEM TESTING

Philosophy:

For system testing, the entire secure wireless data-loading pipeline is demonstrated as a single black-box system and is validated against high-level functionality and security requirements. Our system that is being tested includes the data-loader Raspberry Pi running our C++ ARINC615A application, the authentication process, the wireless link between the two processes, an aircraft load target Pi, and the wireless access point. We are focusing on testing each pipeline individually and once we have working components they can be connected and integrated together and testing the overall system. It is important to test a secure data load by validating users properly and test the rejection of unauthorized clients. We also need to test the behavior under network connectivity issues and be able to test the speed of Wi-Fi during transmission. We need to test and document the performance and stability of data loading speeds to give an idea of how compatible our solution is. We need to make sure we are demonstrating test cases for the complete system – not just individual components – to verify we are meeting the requirements of security, reliability, and performance.

Tools:

To implement this approach the first stage is the unit, interface, and integration tests must pass before the system tests can run. We will use CTest to verify our test suites and make sure that the authentication behaves as expected and will make test cases to check if the data can be transferred and received properly. Once these authentication steps are completed, then we can focus on the pipeline testing process and can utilize GitLab CI/CD to automatically run critical end-to-end checks on each of the working pipeline steps.

CTest: Verify test suites. Check for successful data transfers and or data leaks/corruption.

GitLab CI: Automatically run end-to-end checks on each of the working pipeline steps.

5.5 REGRESSION TESTING

Philosophy:

This step will become more important after we have connected all the components to our project, but is important now nonetheless. Testing will be driven by our functional requirements. We will make sure any new addition passes all the old tests. If there is a failure, we will reassess the relevancy of the old tests or the functionality of the new addition. There are some cases where we cannot deem old tests as irrelevant. The critical features for our solution include EAP-TLS, TFTP data load of up to 2 TB, data load between data loader and target, and tracking client certificates.

Tools:

GitLab CI: To ensure all new additions pass regression tests, we will set up GitLab CI pipelines to build and run the tests every commit. This removes human error or leniency from the equation by automating our regression testing.

Docker: We plan to create containerized environments with unique certificate and network configurations to regression test our EAP-TLS implementation. This increases the reproducibility of our solution as well.

CTest: Ensure any new additions to our branch of the ARINC libraries don't break old test cases. Maintains good base functionality.

5.6 ACCEPTANCE TESTING

Philosophy:

We will prototype early and often, planning to demo prototypes to our client every 2-3 weeks. This way we get frequent feedback from our client, and can catch any acceptance failures before making too much progress in the wrong direction. This works with our current meeting schedule, since we already make weekly progress presentations to our client. Two weeks ago we demoed our Authentication prototype for the client, so this testing method is already in effect.

5.7 SECURITY TESTING

Philosophy:

Security testing is an extremely important aspect of our project. Our project is titled "Secure Wireless Aircraft Data Load", and the client has very clear security requirements we must

meet. Because our solution may be used as a standard in avionic technology in the future, we have to do very strenuous security testing.

To make sure we cover all of our test cases, we must individually test each security layer, as well as the solution as a whole. To ensure we have secure channels for the AP to communicate over, EAP-TLS will need to be thoroughly tested. To do this, we will sniff on the channels to try and capture packets, which will then be analyzed in Wireshark. This ensures confidentiality and integrity of the information being sent.

Testing Categories:

1. **Encryption Testing:** Verify all data transmitted over the wireless network remains encrypted under all operational conditions. Testing includes:
 - Capturing raw wireless traffic using monitor mode and verifying that no payload data is readable.
 - Ensuring TLS 1.3 is enforced with strong cipher suites.
 - Attempting downgrade attacks to weaker encryption (expected to fail)
 - Evaluating the strength and correctness of certificate chain validation.

The goal is to confirm that no data, including ARINC packets or metadata, is exposed at any point in transit.

2. **Authentication Testing:** EAP-TLS mutual authentication is central to the system's security. We test:
 - Successful authentication using valid client + server certificates.
 - Rejection of clients with invalid, expired, revoked, or mismatched certificates.
 - Handling of corrupted or tampered certificates.
 - Behavior when CA trust chains are misconfigured.
 - Replay and session-resumption resistance.Authentication must demonstrate that only trusted devices can access the data-loading channel.
3. **Penetration Testing:** We attempt to intentionally disrupt the secure pipeline to ensure resilience.
 - Simulated MITM attacks (rogue AP, rogue client).
 - Deauthentication attacks and testing reconnection behavior.
 - TLS renegotiation attempts.
 - Flooding the AP with bogus EAP messages to ensure system stability.
 - Invalid or malformed ARINC/TFTP request injection.
4. **Certificate Management and Lifecycle Testing:** Given that avionic environments heavily depend on cryptographically trusted identities, we test the full certificate lifecycle:
 - CA creation and rotation
 - Revocation list (CRL) and OCSP behavior
 - Certificate replacement on both client and RADIUS server
 - Handling of expired certificates during load operations

Tools:

Wireshark: For packet capture, inspection, and encryption validation.

FreeRADIUS: For authentication testing, certificate validation, and EAP-TLS behavior analysis.

hostapd / wpa_supplicant: For evaluating AP and client-side wireless authentication behavior.

eapol_test: For automated EAP-TLS verification in scripted test environments. Simulates various kinds of cyber attacks by flooding attack by spamming requests to our RADIUS server.

Valgrind: For memory leak, buffer overflow, and heap safety checks within the ARINC C++ codebase to harden against exploitation.

OpenSSL CLI: For certificate generation, tampering, and negative-case testing.

5.8 USER TESTING

Philosophy:

User testing is tough in our case, since our solution will only be used by aircraft maintenance technicians. We plan to try to get an engineer from Collins Aerospace to take a look at our project solution to determine what flaws there may be. We will need to check with our client on if he is able to get us in contact with someone with those skills.

5.9 RESULTS

So far, our testing has focused on mainly evaluating various network configurations to determine how best to integrate all of the project's components in a way that aligns with Collins Aerospace's operational environment. The initial approach involved hosting the RADIUS server on the Iowa State ETG-provided virtual machine. During this phase, we identified a significant limitation which was that the VM environment did not allow us to appropriately isolate our other devices, which is a requirement for accurately replicating the secure deployment model used by Collins.

To address this, we conducted a series of trial and error using alternative setups. This included configuring the RADIUS server on our Dataloader Raspberry Pi and experimenting with migrating the service from our VM to a dedicated physical server. Each configuration we came up with was assessed for reliability, ease of integration, and its ability to simulate the needed target environment.

Through these iterations, we gained a clearer understanding of how different architectures impact system behavior and security isolation. These findings have helped narrow down which setup best supports accurate testing and long-term functionality for the client. Our next steps involve selecting the optimal configuration and moving forward with more detailed validation once the full system components are integrated.

6 Implementation

Describe any (preliminary) implementations of your design thus far. Support any general, descriptive text with relevant images. If your project has inseparable activities between design and implementation, you can list them either in the Design section or this section.

Our implementation has progressed alongside our iterative design updates. With the following system architecture: including two Raspberry Pi 5 devices, a Cisco 240AC wireless access point, and an Ubuntu 22.04 RADIUS authentication server– we have begun constructing a realistic and secure wireless data loaded environment modeled after aircraft data loader workflows. This section outlines the implementation completed to date.

6.1 HARDWARE INTEGRATION

All core hardware components have been acquired and initialized:

- Raspberry Pi 5 (Data Loader / Ground System):
 - Configured to host the ARINC 615A data-loader software. This device simulates the authorized ground maintenance terminal that initiates secure data uploads.
- Cisco Aironet 240AC Wireless Access Point (WAP):
 - Configured as the secure wireless transport layer. Operates as the central radio through which all authenticated communication must pass.
- Ubuntu 22.04 Laptop (RADIUS Server):
 - Acts as the authentication backend using FreeRADIUS, providing WPA2-Enterprise / 802.1X identity verification for all wireless clients.
- Raspberry Pi 5 (Target Aircraft System):
 - Represents the aircraft-side receiving unit for ARINC 615A data transfers

6.2 NETWORK AND AUTHENTICATION IMPLEMENTATION

A major focus of early implementation has been establishing a secure, realistic wireless communication environment using enterprise grade authentication.

RADIUS Server Setup, on the Ubuntu 22.04 laptop:

- Installed and configured FreeRADIUS.
- Created test user/device credentials for EAP authentication.
- Linked FreeRADIUS to the Cisco 240AC WAP via Ethernet for centralized authentication.
- Enabled detailed logging to capture authentication events during test uploads.

Cisco 240AC WAP Configuration, the WAP is configured to emulate an aviation-grade secured wireless network:

- WPA2-Enterprise (802.1X) enabled.
- Authentication pointed to the Ubuntu RADIUS server.
- A dedicated SSID created exclusively for the two Raspberry Pis.
- Separate VLAN used for test isolation (optional but recommended).

This configuration ensures only authenticated devices can join the network, mimicking the security posture required in real aircraft maintenance operations.

6.3 DEVICE CONFIGURATION AND CONNECTIVITY TESTING

Both Raspberry Pi 5 units were configured with identical base software images and wireless settings:

- Installed necessary dependencies for ARINC 615A.
- Joined the WPA2 network through 802.1X authentication.
- Validated correct assignment of IP addresses and routes.

Preliminary connectivity tests have been conducted:

- Verified that the Data Loader Pi can successfully authenticate with RADIUS through the WAP.
- Confirmed the Target Pi also authenticates and remains reachable.
- Validated stable communication between both Pis over the secured wireless network.

6.4 ARINC 615A LOADER INTEGRATION

6.5 END TO END PROTOTYPE WORKFLOW

A basic version of the full workflow:

1. Data Loader Pi attempts to join the secure wireless network.
2. WAP forwards authentication to the RADIUS server.
3. Upon successful authentication, the Data Loader Pi gains network access.
4. Target Pi authenticates similarly and joins the network.
5. Both Pis establish communication over the secure wireless connection.
6. Sample test files have been transferred from Data Loader Pi → Target Pi to validate the pipeline

6.6 SUMMARY

The revised system architecture has been successfully implemented at a foundational level. The RADIUS-authenticated wireless environment is operational, both Raspberry Pis successfully connect through enterprise-grade authentication, and initial connectivity tests confirm readiness for deeper ARINC 615A integration. These early implementation steps provide a stable platform upon which the secure wireless data-loading prototype will be completed.

7 Ethics and Professional Responsibility

Use this section to describe your considerations of engineering ethics and professional responsibility. Most importantly how are you defining engineering ethics and professional responsibility in the context of your project and what steps are you taken to ensure ethical and responsible conduct. Each section references one type of ethical/professional responsibility considerations. You may also use this introductory section to note any overarching ethical philosophy among your team.

7.1 AREAS OF PROFESSIONAL RESPONSIBILITY/CODES OF ETHICS

This discussion is with respect to the paper by J. McCormack and colleagues titled “Contextualizing Professionalism in Capstone Projects Using the IDEALS Professional Responsibility Assessment”, *International Journal of Engineering Education* Vol. 28, No. 2, pp. 416–424, 2012

Pick one of IEEE, ACM, or SE code of ethics (all linked in class slides). Create a table, like Table 1 in the McCormack et al. (2012, pg. 418)) paper, with the following columns representing: Area of Responsibility (from the paper), Definition (in your own words), Relevant Item from Code of Ethics (from the Code of Ethics you selected, and description of how your team has interacted with that area of professional responsibility or adhered to that code during your project thus far.

In text below the table, describe one area in which your team is performing well. Describe what your team is doing and how that signifies strong performance. Also describe one area in which your team needs to improve. Describe what your team is currently doing and what it should do in the future to improve.

Table X. [ADD DESCRIPTION]

Area of responsibility	Definition	NSPE Canon
High Performance	[Area that went well]	

7.2 FOUR PRINCIPLES

Create a table with rows for each broader context area (see Section 4.1.1) and columns for each of the four principles (beneficence, non-maleficence, respect for autonomy, and justice; see Beauchamp, 2007). Within the table, identify at least one way each of the four principles applies to each of the broader context areas. Some principle-broader context connections might be more prominent than others, but you should be able to identify something for each table cell. Note: Your design may end up negative or neutral in some cells. For example, your product might perform poorly in environment-nonmaleficence because it utilizes natural resources without a positive/mitigating effect.

Below the table, note one broader context-principle pair that is important to your project. Briefly describe the benefit in that area you are working towards and how you will ensure it. Also note one broader context-principle pair in which your project/end design is or will be lacking. Describe either (a) how this negative is overcome by other positives in other areas of the project/design or (b) what your team must do to improve in this area.

Table 2. The Framework of Principles

Principles	Broader Context-Principle
Respect for autonomy capacities of autonomous persons]	[a principle requiring respect for the decision-making
Nonmaleficence	[a principle requiring not causing harm to others]
Beneficence	[a group of principles requiring that we prevent harm, provide benefits and balance benefits against risks and costs]
Justice	[a group of principles requiring appropriate distribution of benefits, risks and costs fairly]

7.3 VIRTUES

List and define at least three virtues that are important to your team. Describe what you will do or have done as a team to support these virtues among all team members.

Each team member should also answer the following:

- Identify one virtue you have demonstrated in your senior design work thus far? (Individual)
 - o Why is it important to you?
 - o How have you demonstrated it?
- Identify one virtue that is important to you that you have not demonstrated in your senior design work thus far? (Individual)
 - o Why is it important to you?
 - o What might you do to demonstrate that virtue?

[Paragraph describing our core virtues]

8 Closing Material

8.1 CONCLUSION

Summarize the work you have done so far. Briefly re-iterate your goals. Then, re-iterate the best plan of action (or solution) to achieving your goals. What constrained you from achieving these goals (if something did)? What could be done differently in a future design/implementation iteration to achieve these goals?

8.2 REFERENCES

List technical references and related work / market survey references. Do professional citation style (ex. IEEE). See link:

<https://iee-dataport.org/sites/default/files/analysis/27/IEEE%20Citation%20Guidelines.pdf>

8.3 APPENDICES

Any additional information that would be helpful to the evaluation of your design document.

If you have any large graphs, tables, or similar data that does not directly pertain to the problem but helps support it, include it here. This would also be a good area to include hardware/software manuals used. May include CAD files, circuit schematics, layout etc., PCB testing issues etc., Software bugs etc.

9 Team

Complete each section as completely and concisely as possible. We strongly recommend using tables or bulleted lists when applicable.

9.1 TEAM MEMBERS

- | | |
|--------------------|----------------------|
| 1) Gurumanie Singh | 2) Anish Nag |
| 3) Declan Creadon | 4) Justin Rohde |
| 5) Doyle Chism | 6) Lincoln Hatlestad |

9.2 REQUIRED SKILL SETS FOR YOUR PROJECT

(if feasible – tie them to the requirements)

9.3 SKILL SETS COVERED BY THE TEAM

(for each skill, state which team member(s) cover it)

9.4 PROJECT MANAGEMENT STYLE ADOPTED BY THE TEAM

Typically, Waterfall or Agile for project management.

9.5 INITIAL PROJECT MANAGEMENT ROLES

(Enumerate which team member plays what role)

9.6 TEAM CONTRACT

Team Name: sdmay26-projo36

Team Members:

- | | |
|--------------------|----------------------|
| 1) Gurumanie Singh | 2) Anish Nag |
| 3) Declan Creadon | 4) Justin Rohde |
| 5) Doyle Chism | 6) Lincoln Hatlestad |

Team Procedures

- Day, time, and location (face-to-face or virtual) for regular team meetings:

Group Meeting: Tues. 1:00-2:00 PM, Library

Client Group Meeting: Thurs. 3:30 PM, Teams

Advisor Group Meeting: (Initial) Thurs, 1:40 PM, In-person w/ Professor Selim

- Preferred method of communication updates, reminders, issues, and scheduling (e.g., e-mail, phone, app, face-to-face):

In-person is preferred for team meetings. Project updates within our team will be made with MS Teams or our message group chat. Client and faculty advisor emails will be shared with the team before going out, ensuring high professionalism.

- Decision-making policy (e.g., consensus, majority vote):

Majority Vote

- Procedures for record keeping (i.e., who will keep meeting minutes, how will minutes be shared/archived):

Document Notes and have a plan for week-to-week requirements (Agile Workflow)

Booking Meeting Spaces: Lincoln Hatlestad

Meeting Minutes: Lincoln Hatlestad

Meeting minutes will be taken in MS OneNote and shared within our Teams chat after meetings.

Participation Expectations

1. Expected individual attendance, punctuality, and participation at all team meetings:

Give a heads up an hour before the meeting if delayed/unable to attend.

At a minimum, four out of six members should be present, and one person from each pair must be present at the meeting. If not, reschedule the meeting.

2. Expected level of responsibility for fulfilling team assignments, timelines, and deadlines:

Each team member will be paired with another, and they will keep each other accountable.

3. Expected level of communication with other team members:

Response time < 6hrs on either Teams or iMessage.

4. Expected level of commitment to team decisions and tasks:

Expecting team members to skip club and/or social events if a high-priority task will miss its deadline otherwise. Ideally, team members work hard to stay ahead of deadlines, so no sacrifices to personal time are needed.

Leadership

1. Leadership roles for each team member (e.g., team organization, client interaction, individual component design, testing, etc.):

Gurumanie Singh Dhiman: Client Interaction, PKI Infrastructure Design

Anish Nag: Protocols/Standards Security Research

Justin Rohde: Infrastructure Development, Dev Board Assembly, Security Testing

Doyle Chism: PKI Research and Implementation

Declan Creadon: Wireless Component Testing

Lincoln Hatlestad: Team Organization, Software Testing

2. Strategies for supporting and guiding the work of all team members:

Buddy system to ensure there is at least one person in the group who is tracking and noting down the work done by another team member. (Pair work)

Weekly reports are done by each pair to update the team on the tasks in progress/completed.

3. Strategies for recognizing the contributions of all team members:

During weekly meetings, we will set expectations for the next meeting to ensure we are all contributing and completing all tasks we set for ourselves and Daniel(Collins).

Collaboration and Inclusion

1. Describe the skills, expertise, and unique perspectives each team member brings to the team.

Anish Nag - embedded security

Justin Rohde - Digital Forensics, Reverse engineering, Linux, Python

Doyle Chism - Backend Development, Linux, Python, Generative AI

Lincoln Hatlestad - Frontend Development, Customer Service background

Gurumanie Singh Dhiman - Wireless Networks, Cyber, Linux, Python

Declan Creadon

2. Strategies for encouraging and supporting contributions and ideas from all team members:

Make it necessary for everyone to contribute one unique idea that another team member has not come up with yet (no matter how good or bad).

3. Procedures for identifying and resolving collaboration or inclusion issues (e.g., how will a team member inform the team that the team environment is obstructing their opportunity or ability to contribute?)

Speak with the advisor (Prof Selim) on such matters if they are unable to be resolved within the group.

Goal-Setting, Planning, and Execution

1. Team goals for this semester:

Ensure we complete our research on Wireless Network Security and Aircraft data/guidelines to ensure we can quickly transition from planning to execution.

2. Strategies for planning and assigning individual and team work:

Use a shared OneNote in Teams to assign tasks to each team member and pair.

3. Strategies for keeping on task:

Keep tasks and deadlines within a week to minimize overloading of tasks and ensure everyone stays on track.

Consequences for Not Adhering to Team Contract

1. How will you handle infractions of any of the obligations of this team contract?

Speak to our advisor (Prof Selim) or Professors (Nate and Diane).

2. What will your team do if the infractions continue?

Swap some members around with different partners within the group.

a) *I participated in formulating the standards, roles, and procedures as stated in this contract.*

b) *I understand that I am obligated to abide by these terms and conditions.*

c) *I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.*

1) Gurumanie Singh Dhiman

DATE: 09/02/25

2) Doyle Chism

DATE: 09/09/2025

3) Lincoln Hatlestad

DATE: 09/16/2025

4) Declan Creadon

DATE: 09/16/2025

5) Justin Rohde

DATE: 09/16/2025

6) Anish Nag

DATE: 09/16/2025